

Vol 10 Chapter 12 — Numerical Methods and Computation

Keeper (author pass — deep math/physics revision)

2026-05-24 Sunday

Chapter 12 — Numerical Methods and Computation

Level 1 — one sentence

Numerical methods convert analytic problems into computations (ODE/PDE solvers, eigenvalue routines, Monte Carlo integration, FFTs, finite-difference/finite-element methods) — and the BST team's ~3500 toys are computational verifications using this apparatus, all archived in the `play/` directory of the BST repository.

Level 2 — graduate-physicist precision

12.1 ODE solvers

- Euler method (forward): simplest; $O(h)$ error per step
- Runge-Kutta 4 (RK4): $O(h^4)$; workhorse
- Adams-Bashforth-Moulton: multistep methods
- Implicit methods: for stiff equations (backward Euler, BDF, Rosenbrock)
- Symplectic integrators for Hamiltonian systems (preserve phase-space volume)

12.2 PDE solvers

- Finite difference: discretize derivatives; explicit/implicit/Crank-Nicolson schemes
- Finite element: weak form, basis functions on mesh
- Spectral methods: Fourier or Chebyshev basis, exponential convergence for smooth problems
- Multigrid: hierarchical relaxation for elliptic problems
- Discontinuous Galerkin: handles shocks, complex geometries

12.3 Linear algebra

- LU decomposition for solving $Ax = b$
- QR decomposition for least-squares, eigenvalues
- SVD for general matrix factorization
- Lanczos / Arnoldi for large sparse eigenvalue problems

Standard libraries: NumPy/SciPy, LAPACK, MAGMA (GPU), PETSc, Trilinos.

12.4 Monte Carlo methods

- Plain MC: $\int f d\mu \approx (1/N) \sum f(x_i)$ with x_i random samples; error $\sim 1/\sqrt{N}$
- Importance sampling: weight by good distribution
- Markov chain MC: Metropolis-Hastings, Hamiltonian MC for Bayesian inference
- Quantum MC: path integral, world-line, lattice gauge theory

12.5 FFT

Fast Fourier Transform: $O(N \log N)$ vs naive $O(N^2)$. Foundation for signal processing, PDE solvers, convolution, polynomial multiplication. Cooley-Tukey 1965.

12.6 BST toy practice

The BST team's ~3500 toys in `play/` directory implement these numerical methods to verify specific BST predictions. Pattern: 1. State claim (docstring) 2. Define BST quantities (BST primaries, formulas) 3. Evaluate numerically 4. Compare to observation (target precision stated) 5. PASS/FAIL with explicit numerical precision 6. SCORE line at end

Casey's standing rule: every claim has a toy. Build computational evidence simultaneously with theoretical derivation.

12.7 K-audit anchors

- BST `play/README.md`: toy verification framework
- `play/claim_number.sh`: atomic toy-number assignment
- 2056+ toys as of recent counts

Level 3 — 5th-grader accessibility

Numerical methods = computer methods for physics. ODE solvers (RK4), PDE solvers (finite element), linear algebra (LAPACK), Monte Carlo, FFT — the standard toolkit. BST practice: every claim has a “toy” — a short Python script that verifies the claim numerically. ~3500 toys exist. No theorem without computational evidence. Casey's standing rule: “/toy claim before write.”

Vol 10 closing

This volume developed the math toolkit for the BST physics volumes. The math foundations: Hilbert spaces (Ch 1), complex analysis (Ch 2), ODEs + PDEs (Ch 3-4), Fourier (Ch 5), group theory + Lie groups (Ch 6, LOAD-BEARING), differential geometry (Ch 7, LOAD-BEARING — Bergman curvature → Newton's G), representation theory (Ch 8, LOAD-BEARING — Wallach), special functions (Ch 9), variational + path integrals (Ch 10), asymptotics + WKB (Ch 11), numerical methods (Ch 12).

Vol 11 (Generative Geometry + Topology) extends these foundations with the BST-specific math: K3 surfaces, Heegner numbers, Monster moonshine, Mersenne primes, Cremona elliptic curves, generative-geometry framing.

Where to look this up

- Press et al., *Numerical Recipes*
- BST `play/README.md`